Paper details	Dataset	Application area	Method	Results	Strengths and limitations	Research gap and relation to our work
Binary and Multi-Class Malware Threads Classification Ismail Taha Ahmed; Norziana Jamil; Marina Md. Din; Baraa Tareq Hammad **Year: 2022**	MaleVis: 14,226 RGB byte-images; 26 classes (25 malware families + 1 cleanware); 9,100 training and 5,126 testing images	Static, image-based malware detection and family classification.	RGB-to-grayscale conversion; handcrafted SFTA and Gabor texture features; Gaussian Discriminant Analysis (GDA) and Naive Bayes; 10-fold cross-validation.	SFTA-GDA: 97% binary detection accuracy and about 98% multiclass accuracy. Gabor-GDA 93%; SFTA-NB 84%; Gabor-NB 83% for binary detection	**Strengths:** Low-cost feature vectors; supports binary and multiclass tasks, useful for classical-ML baseline. **Limitations:** One dataset only. Manual feature engineering and no end-to-end learning or behavioral features.	**Gap:** Earlier texture methods used large, costly feature vectors and struggled with similar families. **Relation**: Direct MaleVis baseline for comparing our CNN model on accuracy, automation, and complexity.
A Novel Detection and Multi-Classification Approach for IoT-Malware Using Random Forest Voting of Fine-Tuning CNNs Safa Ben Atitallah; Maha Driss; Iman Almomani **Year: 2022**	MaleVis: 14,226 RGB images; 25 malware classes + 1 benign class; 70% training, 20% validation, 10% testing.	Vision-based IoT malware detection and multiclass family identification.	Transfer learning with ResNet18, MobileNetV2, and DenseNet161; hard voting, soft voting, stacking, and Random Forest-based voting; 5-fold cross-validation.	RF-voting: accuracy 98.68%, precision 98.74%, recall 98.67%, specificity 98.79%, F1 98.70%, MCC 98.65%; 672 ms average processing time.	**Strengths**: Strong ensemble benchmark, broad metrics, transfer learning and compares four fusion strategies. **Limitations:** Single dataset; three-backbone ensemble is resource-heavy and no explainability or runtime behavior.	**Gap:** Single-CNN studies often misclassified similar families and ignored ensemble benefits. **Relation:** Directly uses MaleVis deep-learning benchmark.
Efficient and Generalized Image-Based CNN	Malimg, Microsoft BIG2015, MaleVis, and a blended Malimg+Male	Fast, generalized visual malware multiclass detection	VBDN: malware visualization, BalancedDatasetSampler weighted sampling, image augmentation,	Accuracy: Malimg 94.22%; BIG2015 96.19%; MaleVis	**Strengths:** Four-dataset evaluation; handles imbalance; compact	**Gap:** Existing methods often lacked cross-dataset

Algorithm for Multi-Class Malware Detection Yajun Liu; Hong Fan; Jianguang Zhao; Jianfang Zhang; Xinxin Yin **Year: 2024**	Vis dataset. MaleVis has 9,100 training and 5,126 test images across 26 labels	across heterogeneous datasets.	compact custom CNN, GLCM/classical-ML baselines, and confusion-matrix analysis.	83.22% with the heterogeneous "Other" class and 96.76% after removing it; blended dataset 91.39%	CNN. Detailed class-error analysis. **Limitations:** MaleVis result depends strongly on the other class. Static images omit runtime behavior	generalization, imbalance handling, speed, and class-level error analysis. **Relation:** Guides our sampling, augmentation, lightweight CNN, and confusion-matrix design.
An Enhancement for Image-Based Malware Classification Using Machine Learning with Low Dimension Normalized Input Images Tran The Son; Chando Lee; Hoa Le-Minh; Nauman Aslam; Vuong Cong Dat **Year: 2022**	Malimg: 9,339 images, 25 families; Malheur: 3,133 samples, 24 families. MaleVis is not used	Lightweight visual malware classification for resource-constrained and IoT devices	Horizontal image reduction from 64x64 to 32x64, 16x64, and 8x64; raw pixels fed to k-NN, SVM, and a two-layer CNN; comparison with GIST features.	Malimg: k-NN 98.11% at 8x64; SVM 98.51% at 32x64; CNN about 97.3%. Malheur CNN about 91%. Training time reduced to about 1/2, 1/4, and 1/8.	**Strengths:** Simple dimensionality reduction; retains high accuracy. Also, greatly lowers training cost; useful for low-resource hardware. **Limitations:** Not MaleVis-based.	**Gap:** Square normalization and handcrafted descriptors add unnecessary dimensions and cost. **Relation:** Does not directly realted to MaleVis but supported study that motivates image-resolution and lightweight-preprocessing that helps to understand
An Explainable Hybrid CNN-Transformer Architecture for Visual Malware Classification Mohamme	Training/testing: Malimg, MaleVis, VirusMNIST. External validation: Maldeb and Dumpware-10. MaleVis has 9,100 training and 5,126	Explainable, multi-dataset visual malware classification for SOC/cloud deployment	ConvNeXt-Tiny + Swin Transformer; adaptive gated fusion, auxiliary heads, knowledge distillation, AdamW training, and Grad-CAM explanations	Hybrid average accuracy 94.04%. Malimg 99.57%; MaleVis 88.65% accuracy, 90.74% precision, 96.49%	**Strengths:** Combines local and global features; XAI and multi-dataset testing; external validation. **Limitations**	**Gap:** Few studies combine CNNs and Transformers while remaining explainable and testing across several datasets.

d Alshomrani; Aiiad Albeshri; Abdulaziz A. Alsulami; Badraddin Alturki **Year: 2025**	validation images.			recall, 92.69% F1; VirusMNIST 93.90%; Maldeb 98%; Dumpware-10 97%.	**:** High computational cost and image conversion overhead. MaleVis accuracy trails just simpler baselines.	**Relation:** Closest modern reference for our MaleVis work and a basis for improving efficiency and MaleVis accuracy